

MUHAMMAD AKMAL FIKRI ASMA ZAKI

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PROFESSIONAL SUMMARY

Backend-focused developer experienced in building REST APIs using Spring Boot and FastAPI. Maintain and build a finance portal specialised in helping UTP stakeholders manage customers by reducing manual verification by 60%. Skilled in backend, cybersecurity and database development, with strengths in database analysis and performance optimization

EDUCATION

Universiti Teknologi PETRONAS (UTP), B. CS. (Hons), Seri Iskandar, May 2026 – now

- Year of study: 1st year 1st semester (Current CGPA: 0.00)

Universiti Teknologi MARA (UiTM), Dip. CS., Tapah Campus, September 2023 – Mac 2026 (CGPA:3.78)

PROFESSIONAL EXPERIENCE

Website Developer Intern, **Universiti Teknologi PETRONAS (UTP)**, Seri Iskandar, September 2025 – February 2026

- Using FastAPI and PHP as backend, proposed a financial portal for the UTP Finance department and CEE Study Plan which help **100 Civil Engineering students** plan their academic structure.
- Translated non-technical requirements into technical solutions, collaborating directly with the Senior Executive and Head of Department to complete multiple data analytics tasks and research on civil engineering market.

PROJECT

CEE Study Plan, *Backend Development*

Tech Stack:

- Architected a clean 3-layer REST API using **FastAPI**, **React**, **Supabase** and deployed on **Render** and **Vercel**, seamlessly consumed by a dynamic React frontend, enabling state-driven interactions.
- Developed a full-stack academic planning tool used by **100+** CEE students to monitor course completion and GPA progression in real time while integrated **HuggingFace API** to provide AI feedback on student performance

UTP Vendor Code Requisition (UTP VCR), *Full-Stack Development*

Tech Stack:

- Built and deployed a centralized vendor code requisition portal that cut manual verification workload by **60%** by digitizing a previously email-based approval process.
- Implemented a **3-stage verification** workflow with role separation between standard users and 3 admin tiers to ensure thousands of confidential documents have been verified by respective admin.

GridGuard (AI-assisted traffic incident), *Backend Dev and Project Documentation*

Tech Stack:

- UMHackathon 2026 project that use stateful cyclic reasoning with **LangGraph** loops and tool-driven decisions for budget checks, fund transfer, and contractor dispatch.
- End-to-end auditability in **Postgresql** and operational budgeting in **DuckDB** that **automatically manages budgets** based on AI reasoning that involves thousands to millions of Ringgit project in municipal government.

SKILLS

- **Programming Language:** Python (Intermediate), Java (Intermediate), PHP (Intermediate), Visual Basic (Intermediate), HTML/CSS (Intermediate), JavaScript (Basic), C++ (Basic)
- **Frameworks & Library:** FastAPI (Intermediate), React (Intermediate), Flask (Basic), Spring Boot (Basic)
- **Application Technologies:** Github (Intermediate), Oracle VirtualBox (Intermediate), Burpsuite (Basic), Docker (Basic), Render (Intermediate), Vercel (Intermediate)
- **Database:** PHPMyAdmin MySQL (Intermediate), Supabase Postgresql (Intermediate), Apex Oracle (Basic)
- **Languages:** Natives in Malay, Fluent in English
- **Professional Skills:** Problem-Solving, Critical Thinking, Teamwork

CERTIFICATES

- Vice Chancellor Awards (5x Dean's List) – **Universiti Teknologi MARA (Dip. CS.)**
- Bug Bounty Hunting, How Hackers Find Bugs – **0Day Academy**
- Intro to Github Workshop, React and Copilot Workshop – **Google Developer Student Club (GDSC)**

LEADERSHIP & EXTRA-CURRICULAR ACTIVITIES

Project Director, AWS Workshop

GDSC (Universiti Teknologi PETRONAS) | October 2026

- Direct an organizing committee of 20 members to plan and execute a comprehensive AWS technical workshop for 30 UTP student participants.

Treasurer, Workshop

GDSC (Universiti Teknologi PETRONAS) | September 2026

- Maintain accurate and transparent financial records, ensuring all vendor invoices, equipment rentals, and logistical costs are processed on time.

REFERENCE
